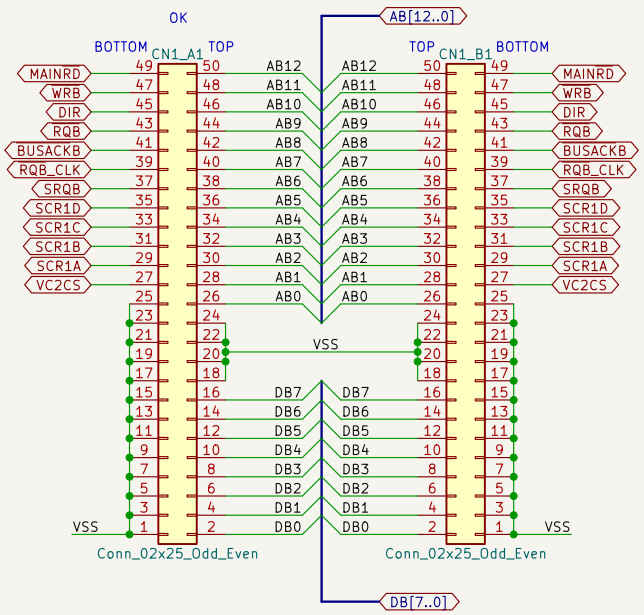
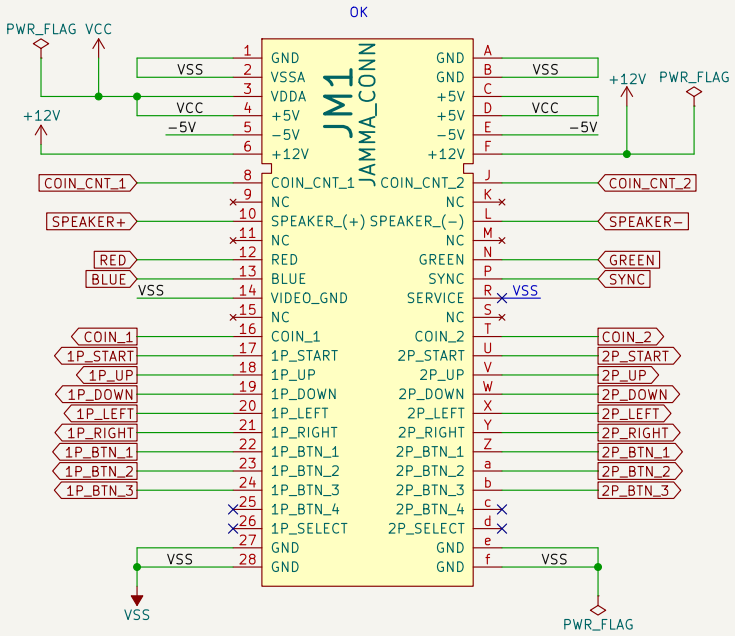
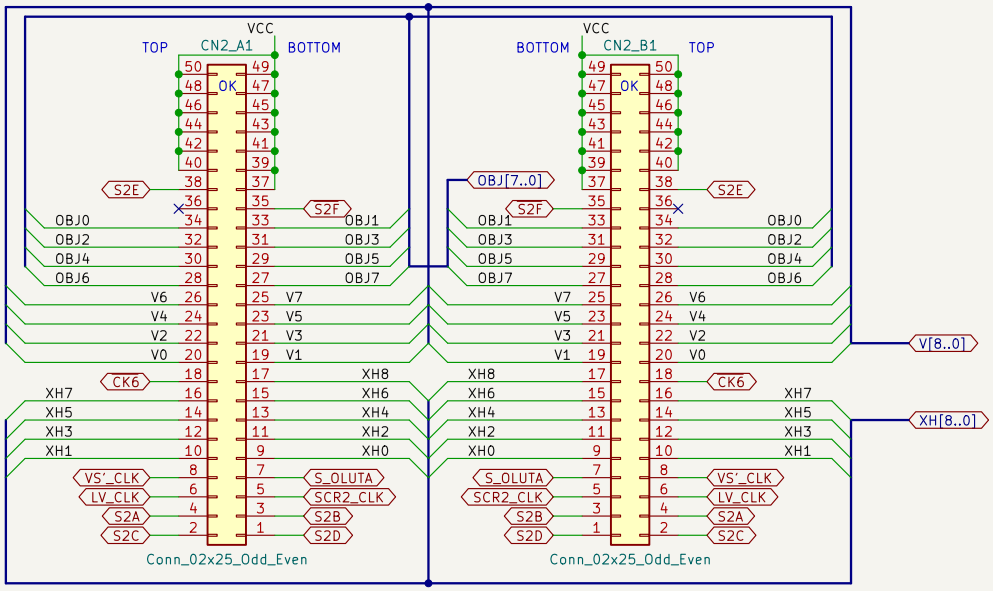






Observation
The pcb that has the jamma connector is the upper level and the location of each device is labeled with number, letter and number1. On the lower level pcb, the devices are labeled with letter and number.

video
File: video.kicad_sch

sound
File: sound.kicad_sch

main
File: main.kicad_sch

interruption
File: interruption.kicad_sch

video_timings
File: video_timings.kicad_sch

v_counter
File: v_counter.kicad_sch

h_counter
File: h_counter.kicad_sch

last_u
File: last_u.kicad_sch

capacitors
File: capacitors.kicad_sch

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JOTEGO

Sheet: /
File: exed.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 1/28

v_scroll

File: v_scroll.kicad_sch

char

File: char.kicad_sch

h_scroll

File: h_scroll1.kicad_sch

object_table

File: object_table.kicad_sch

scroll1

File: scroll1.kicad_sch

obj_ram

File: obj_ram.kicad_sch

scroll2

File: scroll2.kicad_sch

obj_lut

File: obj_lut.kicad_sch

colmix

File: colmix.kicad_sch

obj_counters

File: obj_counters.kicad_sch

logics_gates

File: logics_gates.kicad_sch

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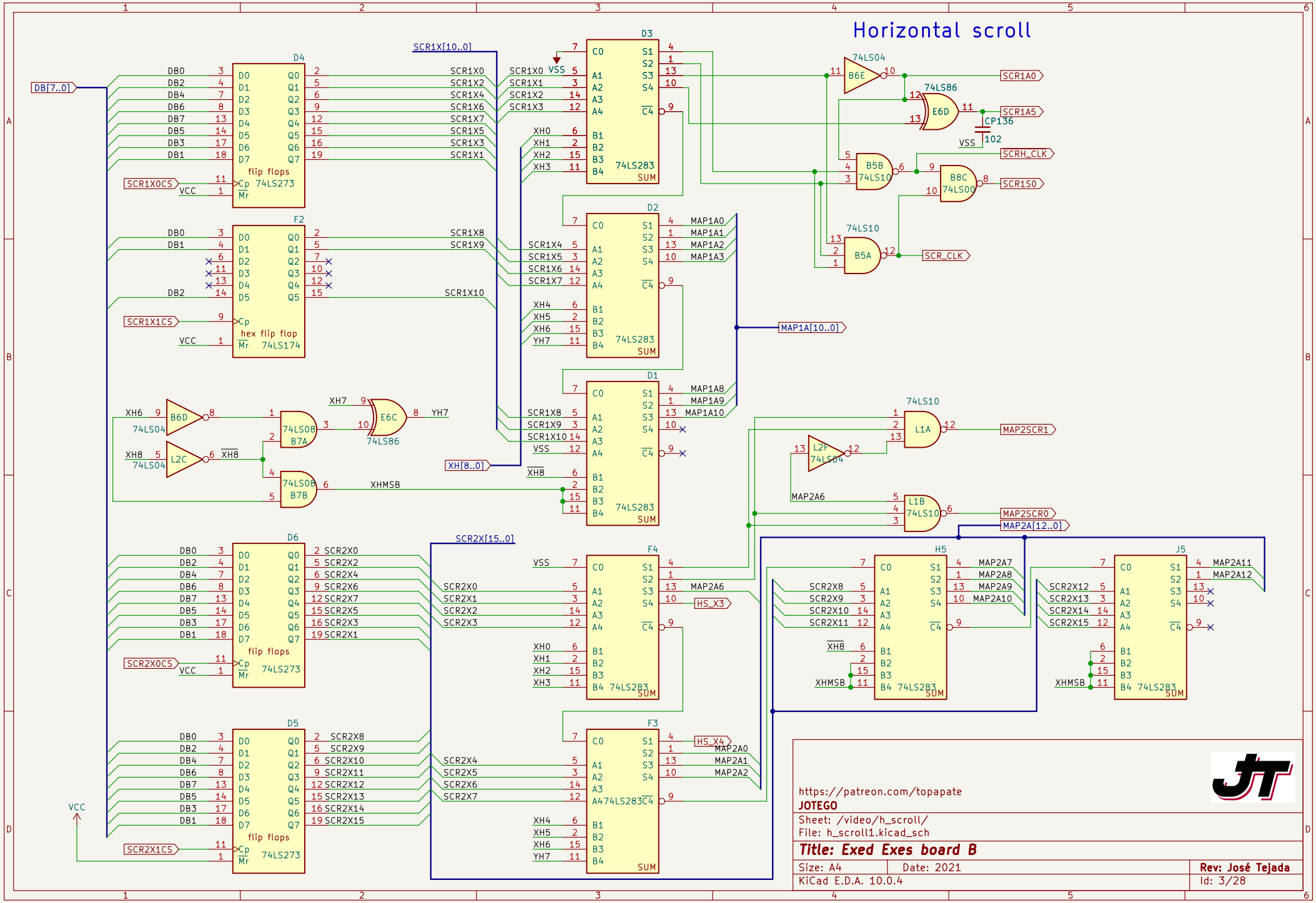
JOTEGO

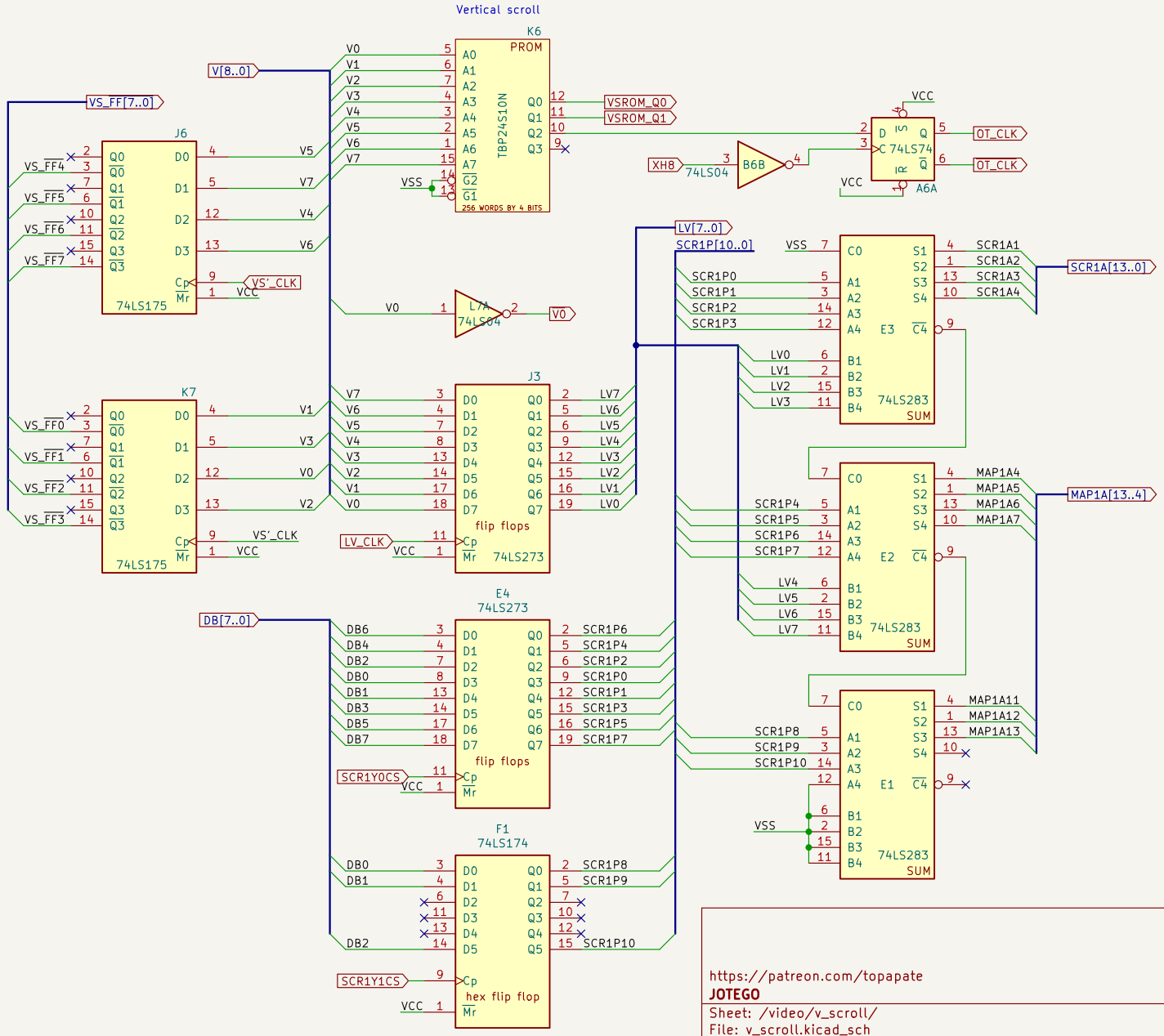
Sheet: /video/

File: video.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 2/28


Horizontal scroll
D4
F2
D6
D5
D3
D2
D1
F4
F3
H5
J5
74LS273
74LS174
74LS283
74LS04
74LS08
74LS86
74LS10
74LS00
flip flops
hex flip flop
SCR1X[10..0]
SCR2X[15..0]
MAP1A[10..0]
MAP2A[12..0]
XH[8..0]
DB[7..0]
VCC
VSS
SCR1X0CS
SCR1X1CS
SCR2X0CS
SCR2X1CS
XHMSB
HS_X3
HS_X4
SCR1A0
SCR1A5
SCRH_CLK
SCR1S0
SCR_CLK
MAP2SCR1
MAP2SCR0
CP136
102
https://patreon.com/topapate
JOTEGO
Sheet: /video/h_scroll/
File: h_scroll1.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 3/28



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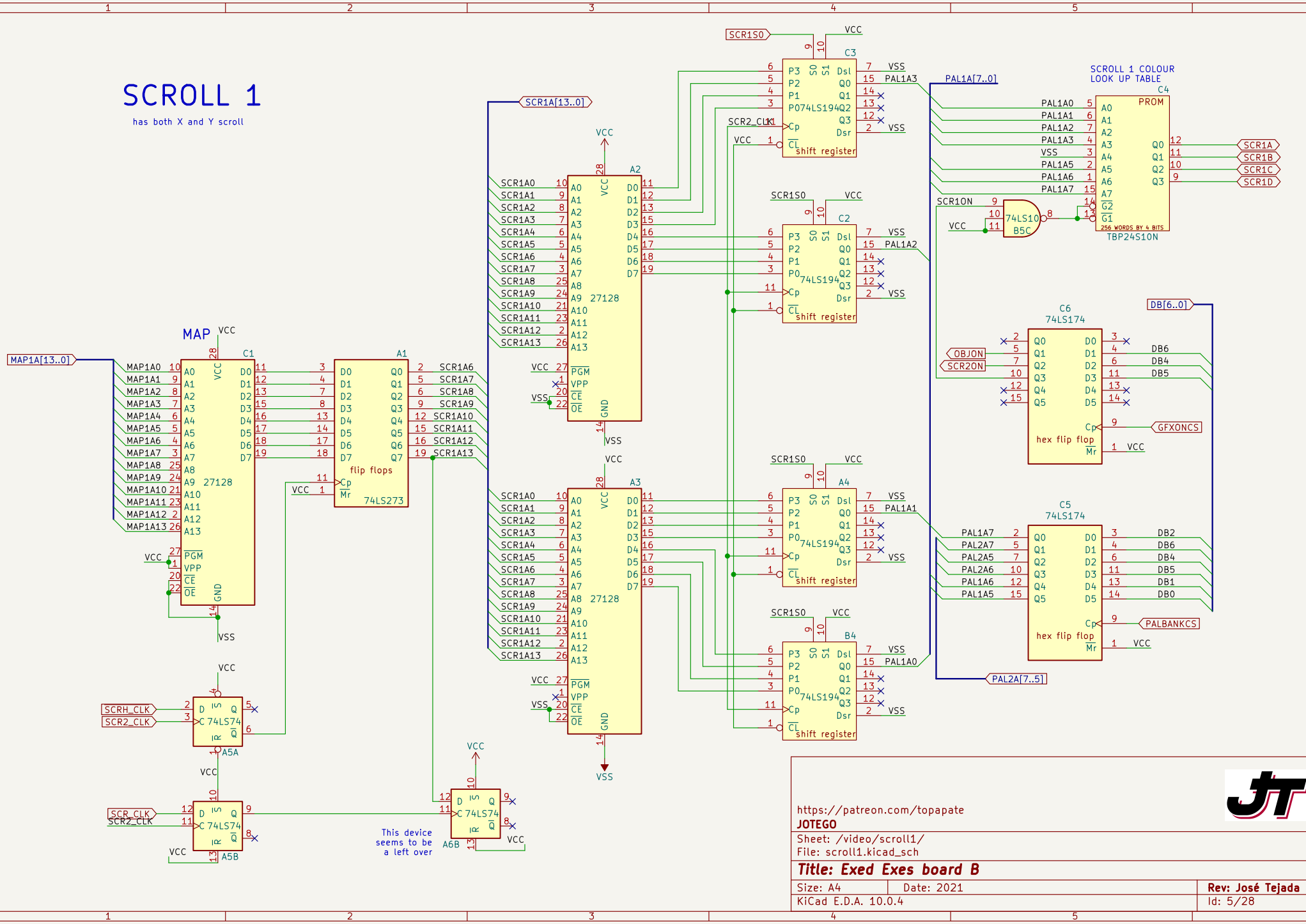
JOTEGO

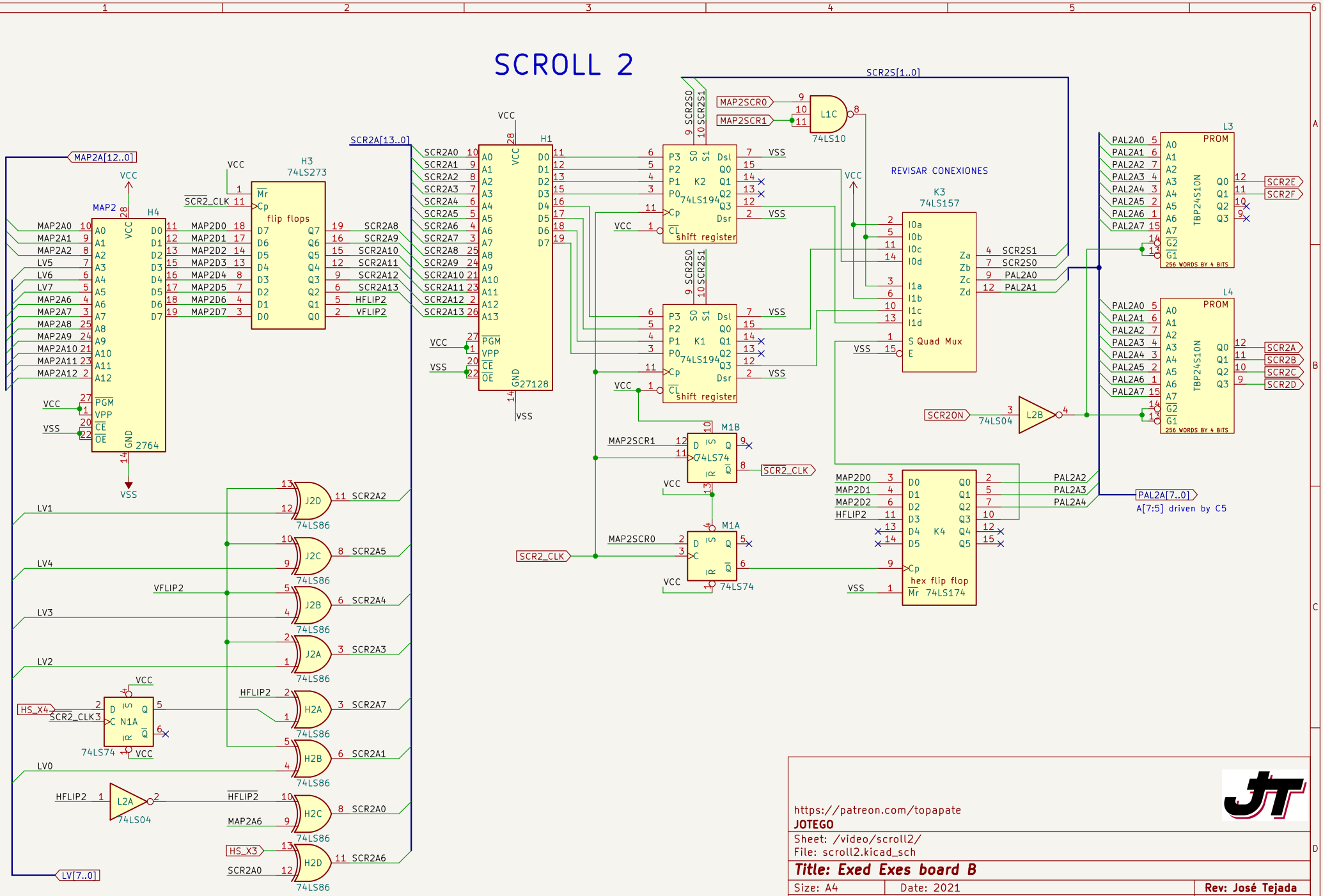
Sheet: /video/v_scroll/

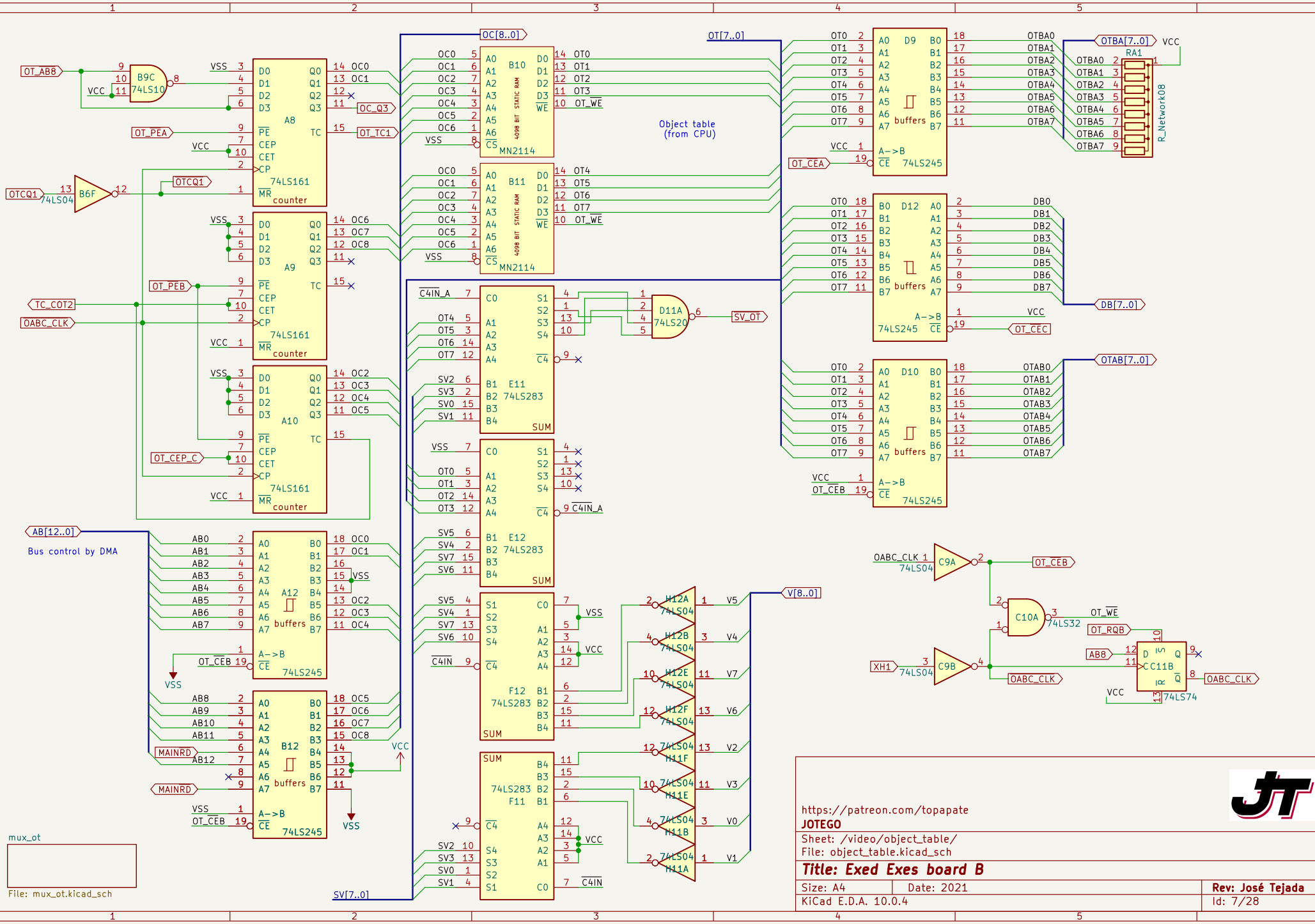
File: v_scroll.kicad_sch

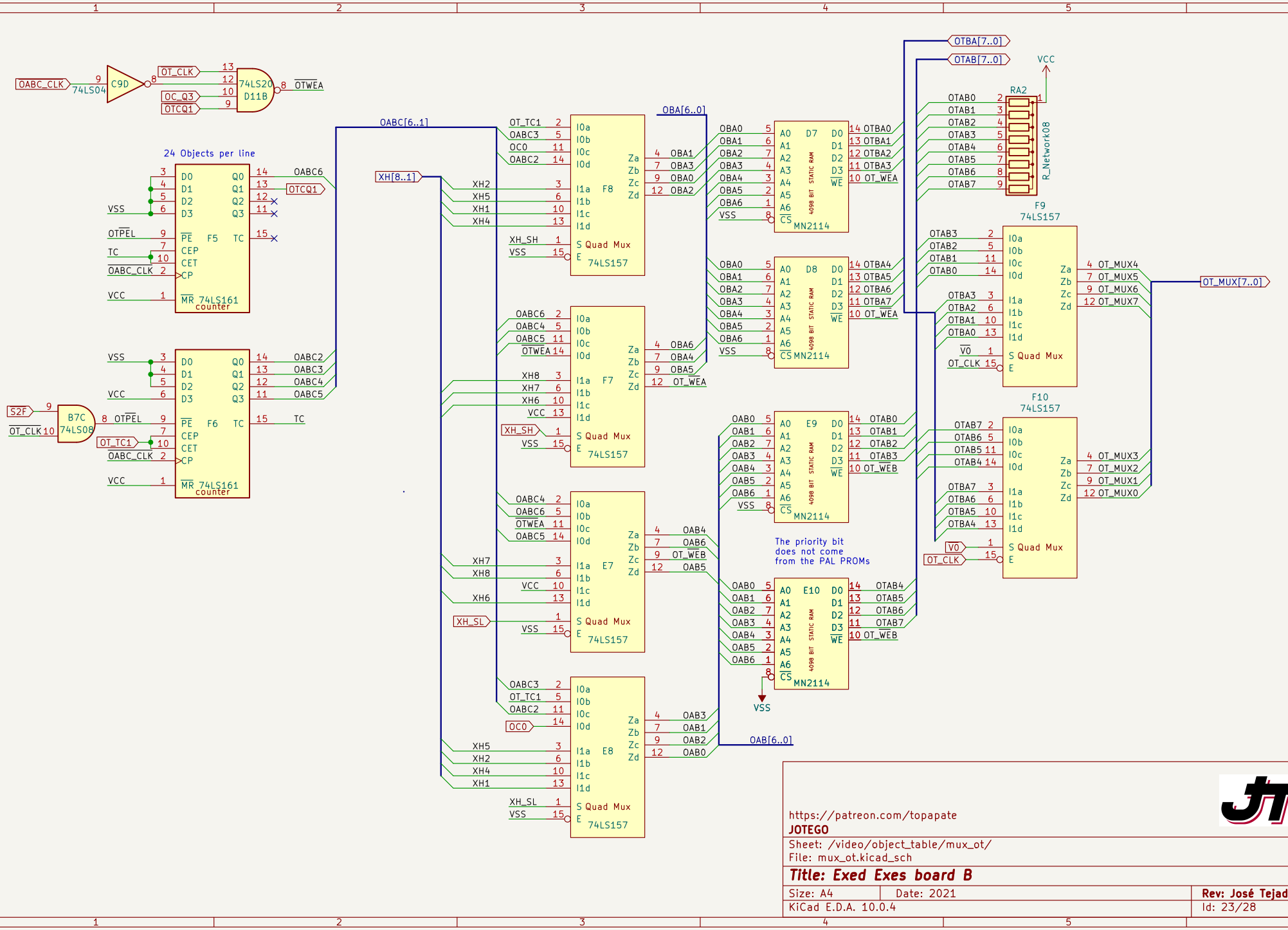
Title: Exed Exes board B

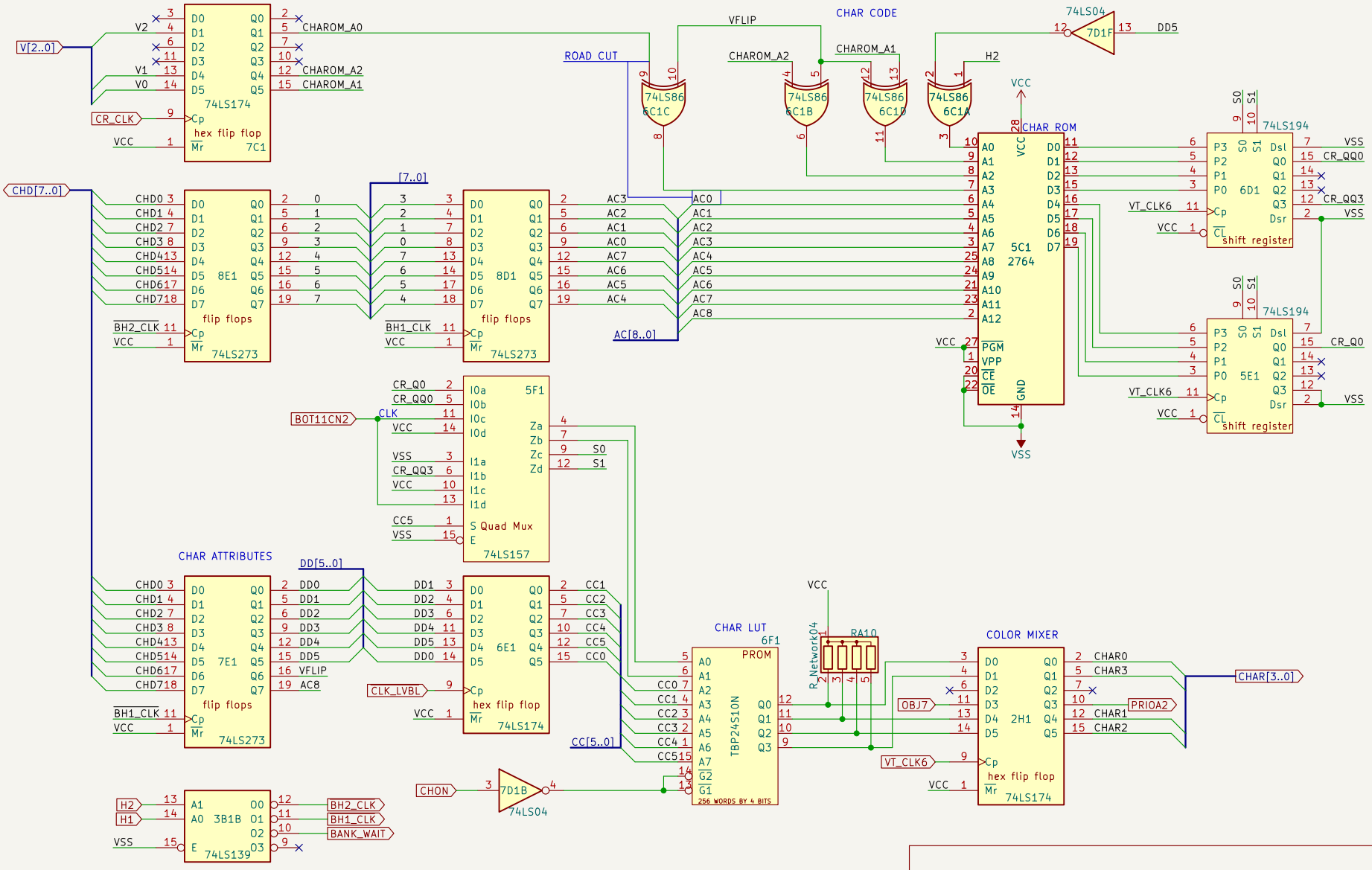
Size: A4	Date: 2021	Rev: José Tejada
KiCad E.D.A. 10.0.4		Id: 4/28


SCROLL 1
has both X and Y scroll
MAP
C1
A1
A2
A3
C2
C3
A4
B4
C4
C5
C6
A5A
A5B
A6B
SCROLL 1 COLOUR
LOOK UP TABLE
PROM
TBP24S10N
74LS74
74LS273
74LS194
74LS10
27128
flip flops
hex flip flop
shift register
MAP1A[13..0]
SCR1A[13..0]
PAL1A[7..0]
PAL2A[7..5]
DB[6..0]
SCR1S0
SCR2_CLK
SCR1ON
OBJON
SCR2ON
GFXONCS
PALBANKCS
SCRH_CLK
SCR_CLK
This device
seems to be
a left over
https://patreon.com/topapate
JOTEGO
Sheet: /video/scroll1/
File: scroll1.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 5/28


SCROLL 2
SCR2S[1..0]
MAP2A[12..0]
SCR2A[13..0]
REVISAR CONEXIONES
MAP2
H4
2764
H3
74LS273
flip flops
H1
27128
K2
K1
74LS194
shift register
K3
74LS157
S Quad Mux
L1C
74LS10
L2B
74LS04
L2A
74LS04
M1B
M1A
74LS74
J2D
J2C
J2B
J2A
H2A
H2B
H2C
H2D
74LS86
K4
hex flip flop
74LS174
L3
L4
PROM
TBP24S10N
256 WORDS BY 4 BITS
LV[7..0]
LV0
LV1
LV2
LV3
LV4
VFLIP2
HFLIP2
SCR2_CLK
HS_X4
HS_X3
MAP2SCR0
MAP2SCR1
SCR2ON
PAL2A[7..0]
A[7:5] driven by C5
https://patreon.com/topapate
JOTEGO
Sheet: /video/scroll2/
File: scroll2.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 6/28


OT_AB8
B9C
74LS10
VCC
VSS
A8
74LS161
counter
OT_PEA
OTCQ1
B6F
74LS04
A9
OT_PEB
TC_COT2
OABC_CLK
A10
OT_CEP_C
AB[12..0]
Bus control by DMA
A12
buffers
74LS245
OT_CEB
MAINRD
B12
VSS
OC[8..0]
B10
B11
4096 BIT STATIC RAM
MN2114
OT[7..0]
Object table
(from CPU)
OT_WE
C4IN_A
D11A
74LS20
SV_OT
E11
E12
74LS283
SUM
C4IN
F12
SV[7..0]
V[8..0]
H12A
H12B
H12E
H12F
H11F
H11E
H11B
H11A
D9
D12
D10
OTBA[7..0]
OTAB[7..0]
DB[7..0]
RA1
R_Network08
OT_CEA
OT_CEC
C9A
C10A
74LS32
C9B
XH1
OT_RQB
C11B
74LS74
mux_ot
File: mux_ot.kicad_sch
https://patreon.com/topapate
JOTEGO
Sheet: /video/object_table/
File: object_table.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 7/28


OABC_CLK
74LS04
C9D
OT_CLK
OC_Q3
OTCQ1
74LS20
D11B
OTWEA
24 Objects per line
OABC[6..1]
XH[8..1]
OBA[6..0]
OABC6
OABC5
OABC4
OABC3
OABC2
OTCQ1
TC
VSS
VCC
OTPEL
S2F
B7C
74LS08
F5
F6
74LS161
counter
F7
F8
F10
74LS157
Quad Mux
MN2114
4098 BIT STATIC RAM
OTBA[7..0]
OTAB[7..0]
RA2
R_Network08
OT_MUX[7..0]
The priority bit
does not come
from the PAL PROMs
OAB[6..0]
https://patreon.com/topapate
JOTEGO
Sheet: /video/object_table/mux_ot/
File: mux_ot.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 23/28



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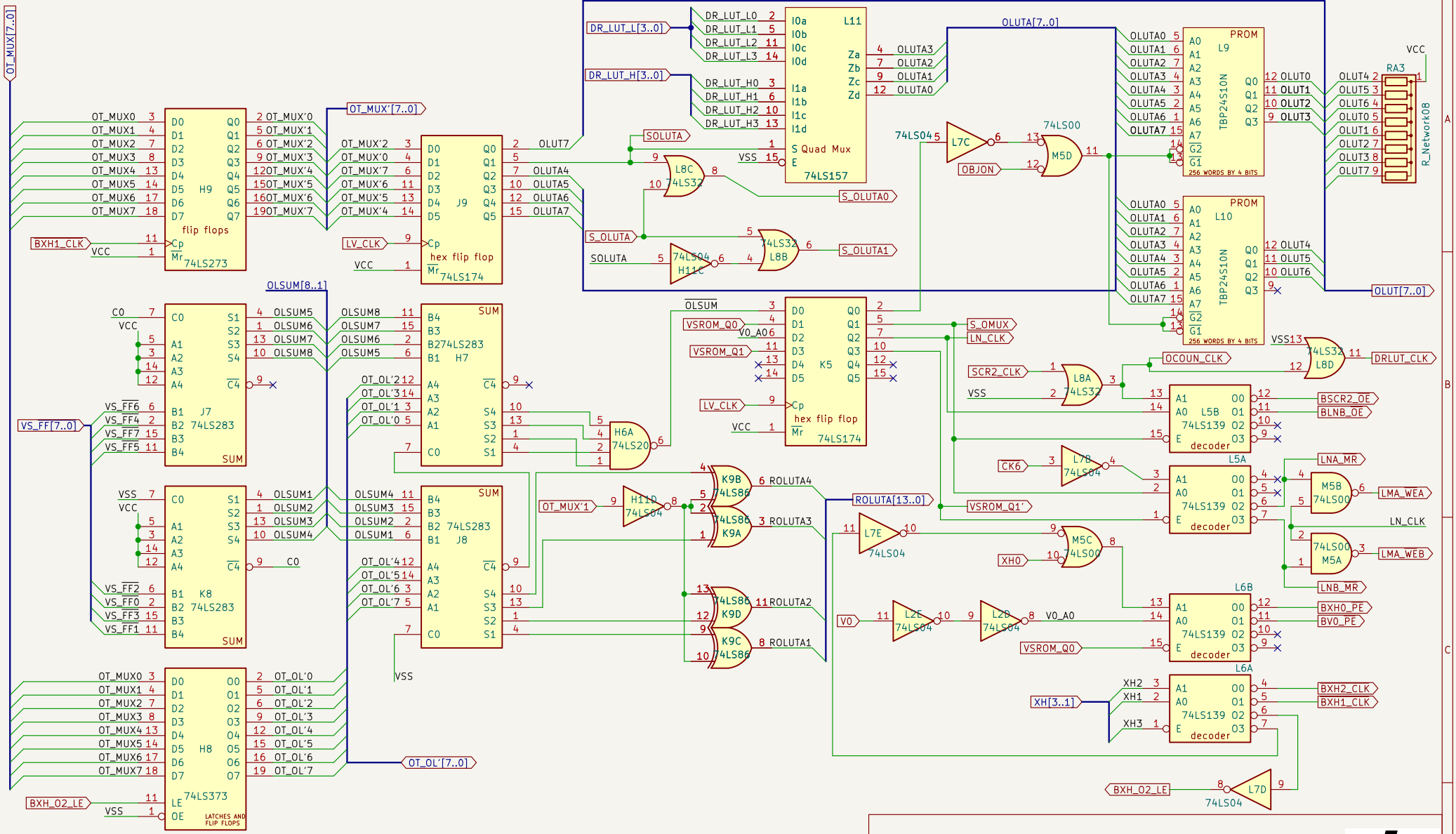
JOTEGO

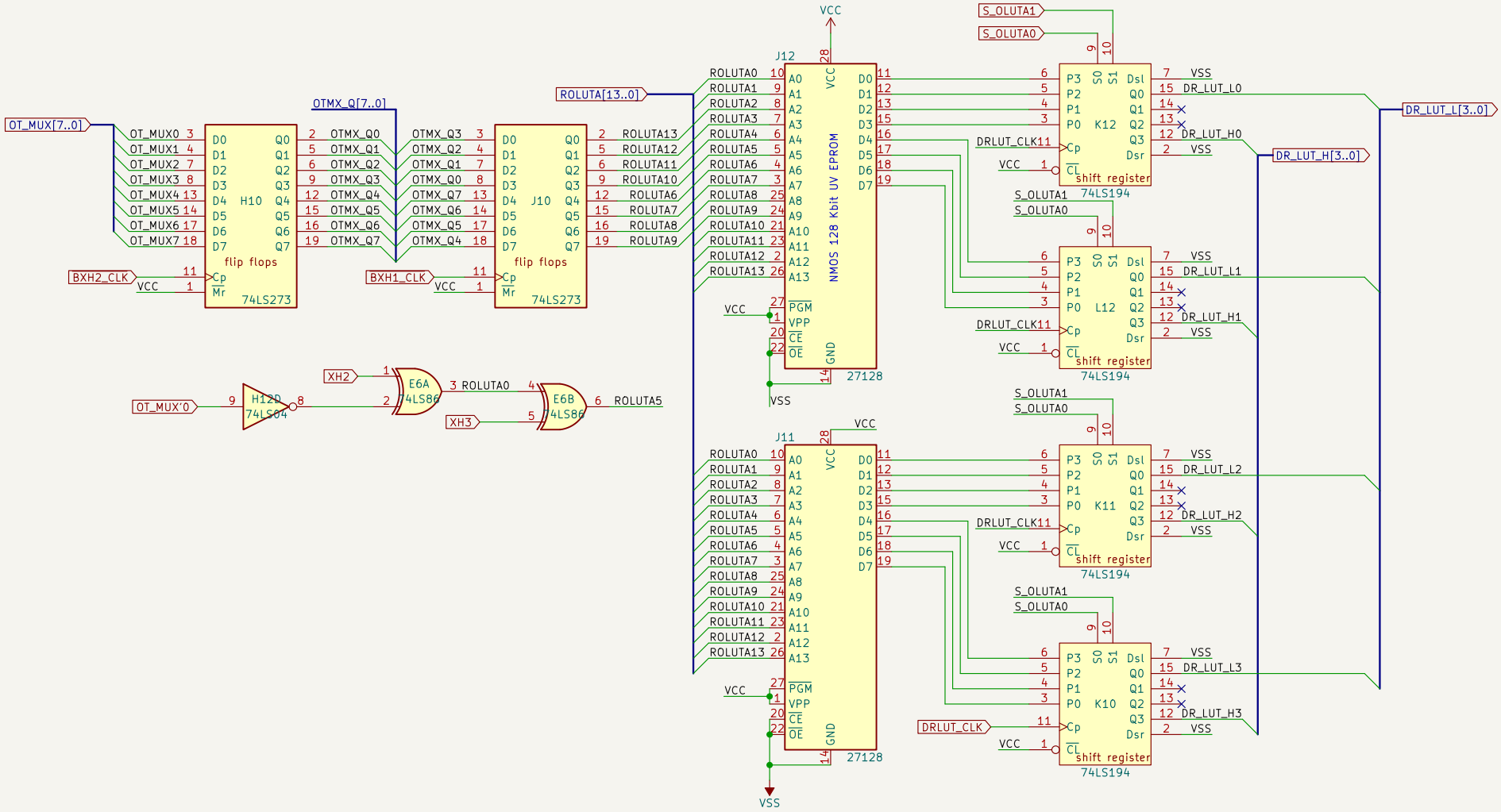
Sheet: /video/char/

File: char.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 8/28


OT_MUX[7..0]
74LS273
flip flops
74LS174
hex flip flop
74LS373
LATCHES AND FLIP FLOPS
OLSUM[8..1]
SUM
74LS283
OT_MUX'[7..0]
OT_OL'[7..0]
VS_FF[7..0]
DR_LUT_L[3..0]
DR_LUT_H[3..0]
74LS157
Quad Mux
OLUTA[7..0]
PROM
TBP24S10N
256 WORDS BY 4 BITS
OLUT[7..0]
R_Network08
74LS139
decoder
74LS04
74LS00
74LS32
74LS86
74LS20
OLSUM
VSROM_Q0
VSROM_Q1
S_OMUX
LN_CLK
OCOUN_CLK
DRLUT_CLK
SCR2_CLK
BSCR2_OE
BLNB_OE
CK6
LNA_MR
LMA_WEA
LMA_WEB
LNB_MR
XH0
XH[3..1]
BXH2_CLK
BXH1_CLK
BXH_02_LE
BXH0_PE
BV0_PE
ROLUTA[13..0]
OBJON
S_OLUTA
SOLUTA
S_OLUTA0
S_OLUTA1
BXH1_CLK
LV_CLK
VCC
VSS
https://patreon.com/topapate
JOTEGO
Sheet: /video/obj_lut/
File: obj_lut.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 9/28
rom_olut
File: rom_olut.kicad_sch



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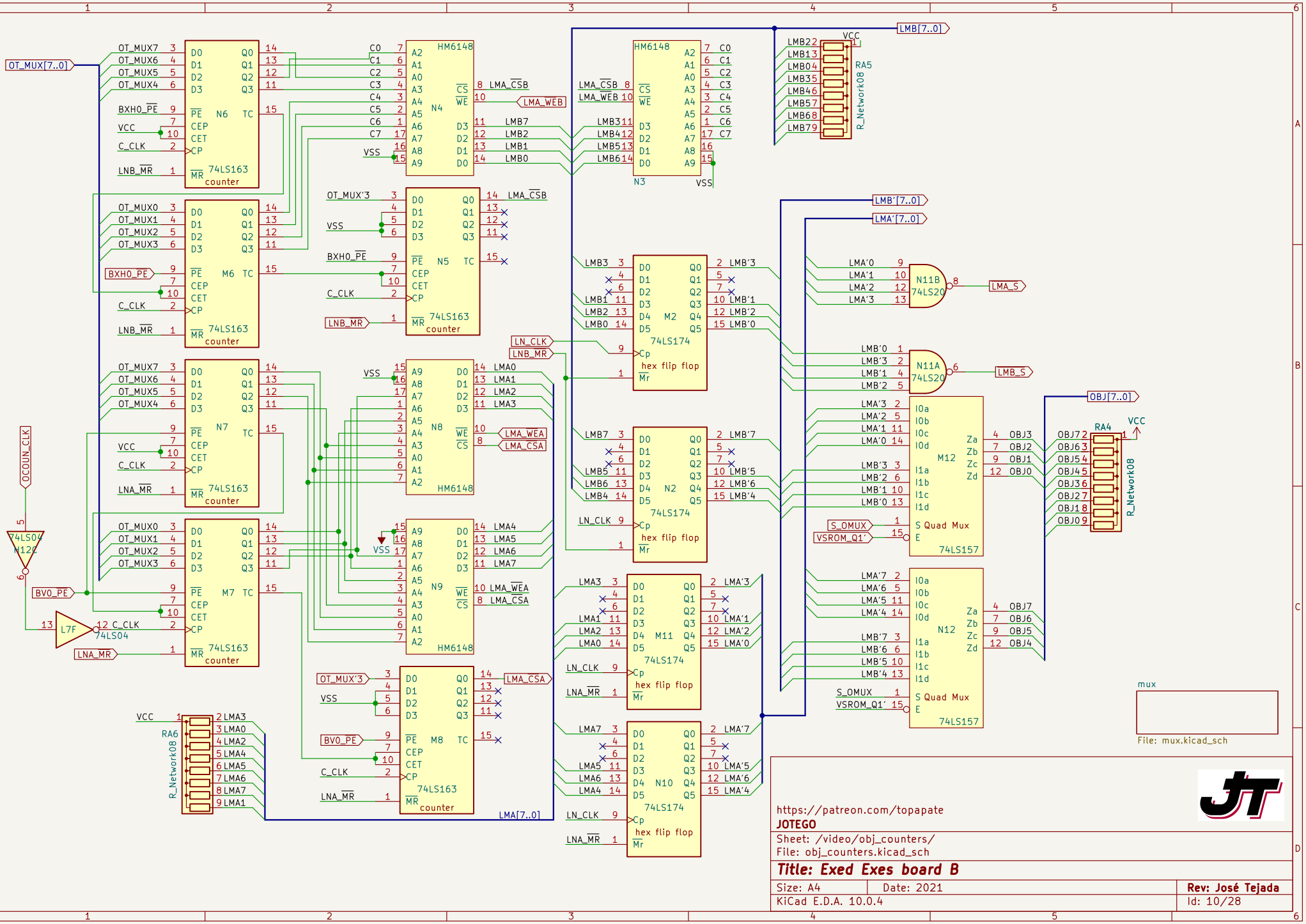
JOTEGO

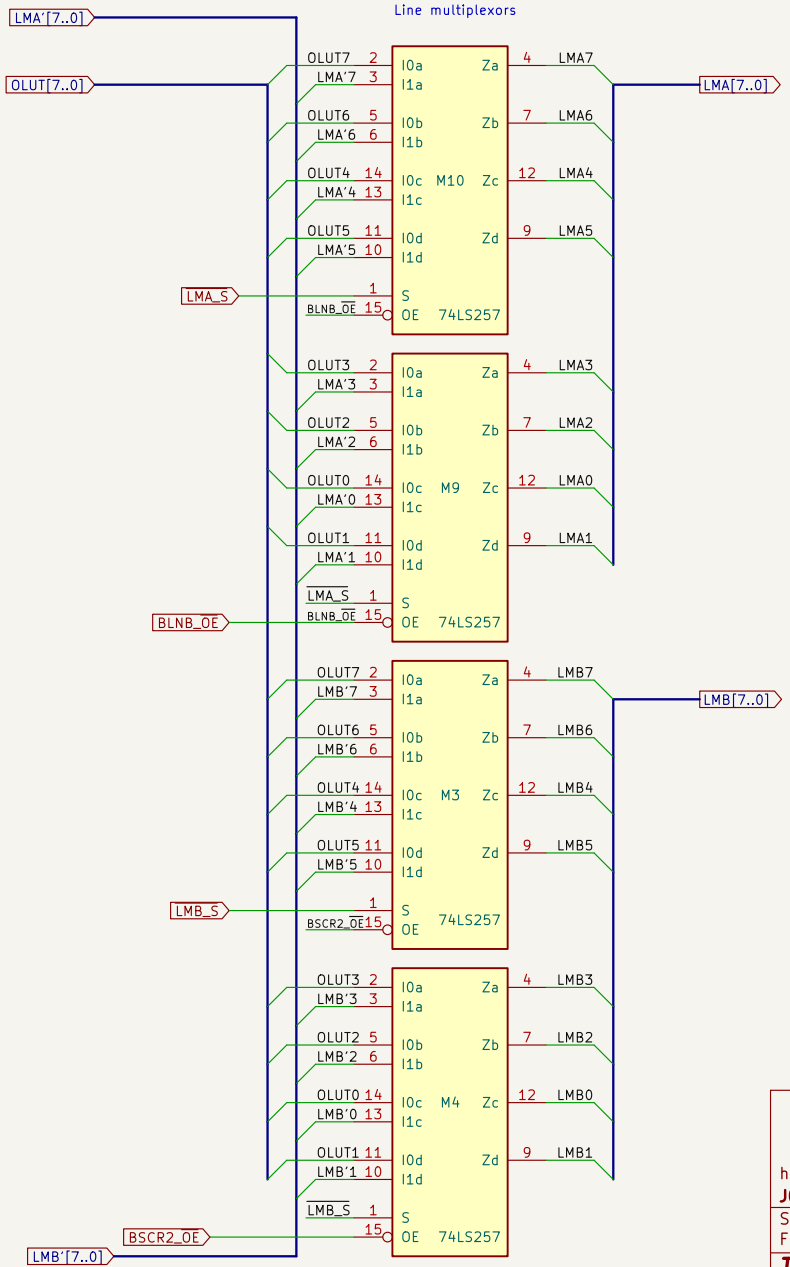
Sheet: /video/obj_lut/rom_olut/

File: rom_olut.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 24/28


OT_MUX[7..0]
OCOUN_CLK
BXH0_PE
BV0_PE
LNB_MR
LNA_MR
LN_CLK
C_CLK
VCC
VSS
LMA_CSB
LMA_WEB
LMA_WEA
LMA_CSA
LMB[7..0]
LMB'[7..0]
LMA'[7..0]
LMA[7..0]
LMA_S
LMB_S
OBJ[7..0]
S_OMUX
VSROM_Q1'
74LS163 counter
74LS174 hex flip flop
74LS157 S Quad Mux
74LS20
74LS04
R_Network08
HM6148
mux
File: mux.kicad_sch
https://patreon.com/topapate
JOTEGO
Sheet: /video/obj_counters/
File: obj_counters.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 10/28



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JOTEGO

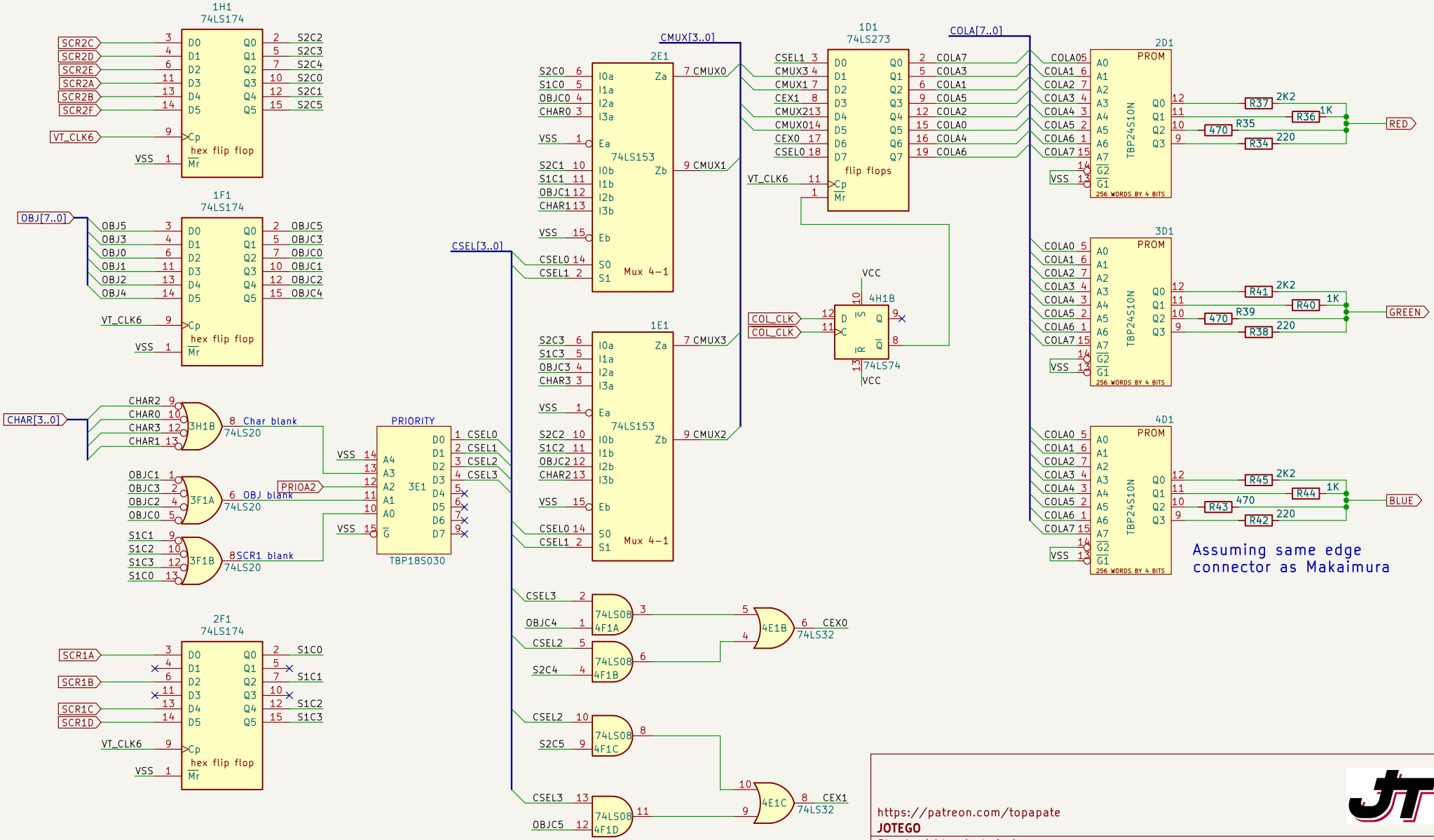
Sheet: /video/obj_counters/mux/

File: mux.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 28/28

COLOR PROMS



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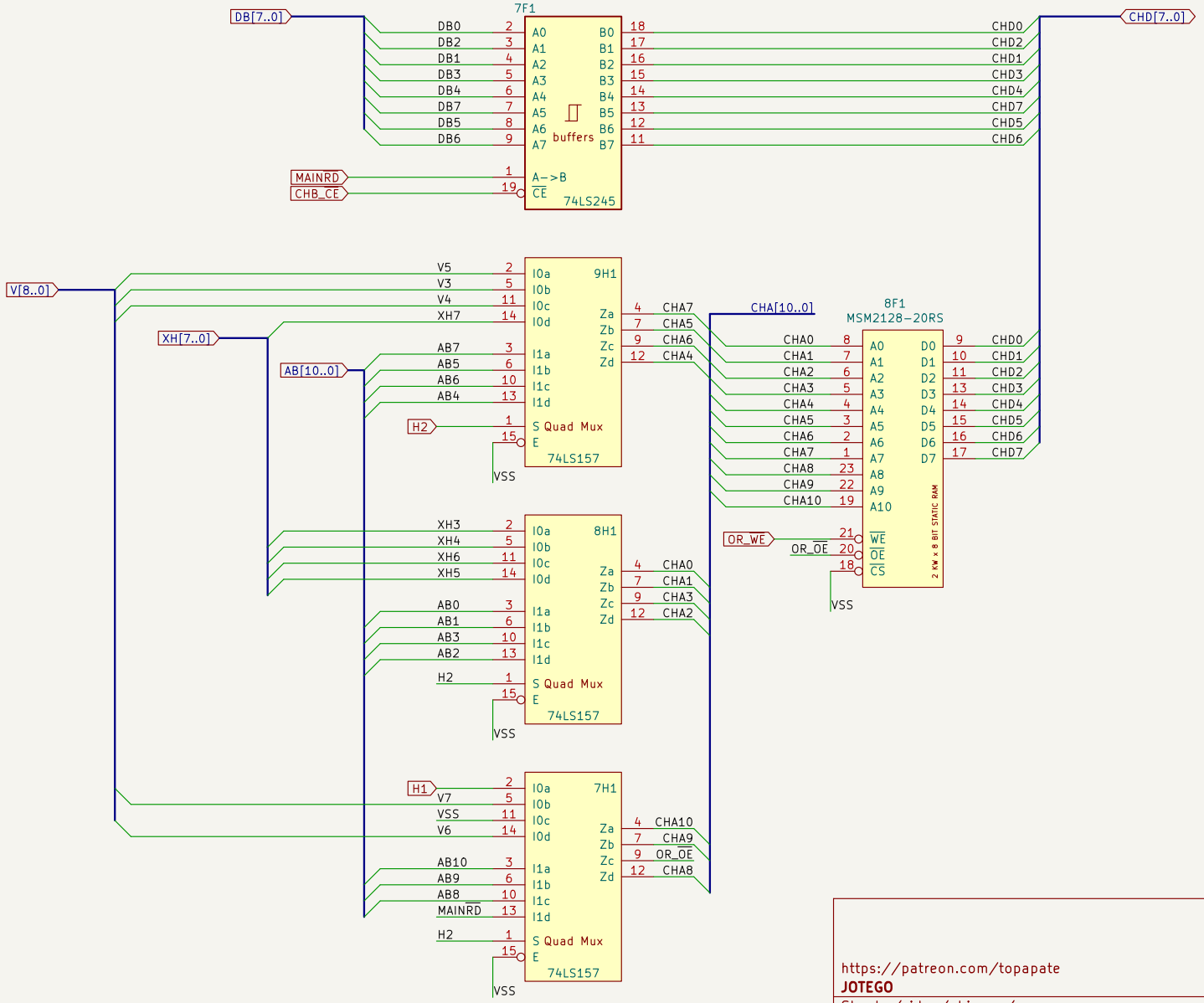
JOTEGO

Sheet: /video/colmix/

File: colmix.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 13/28



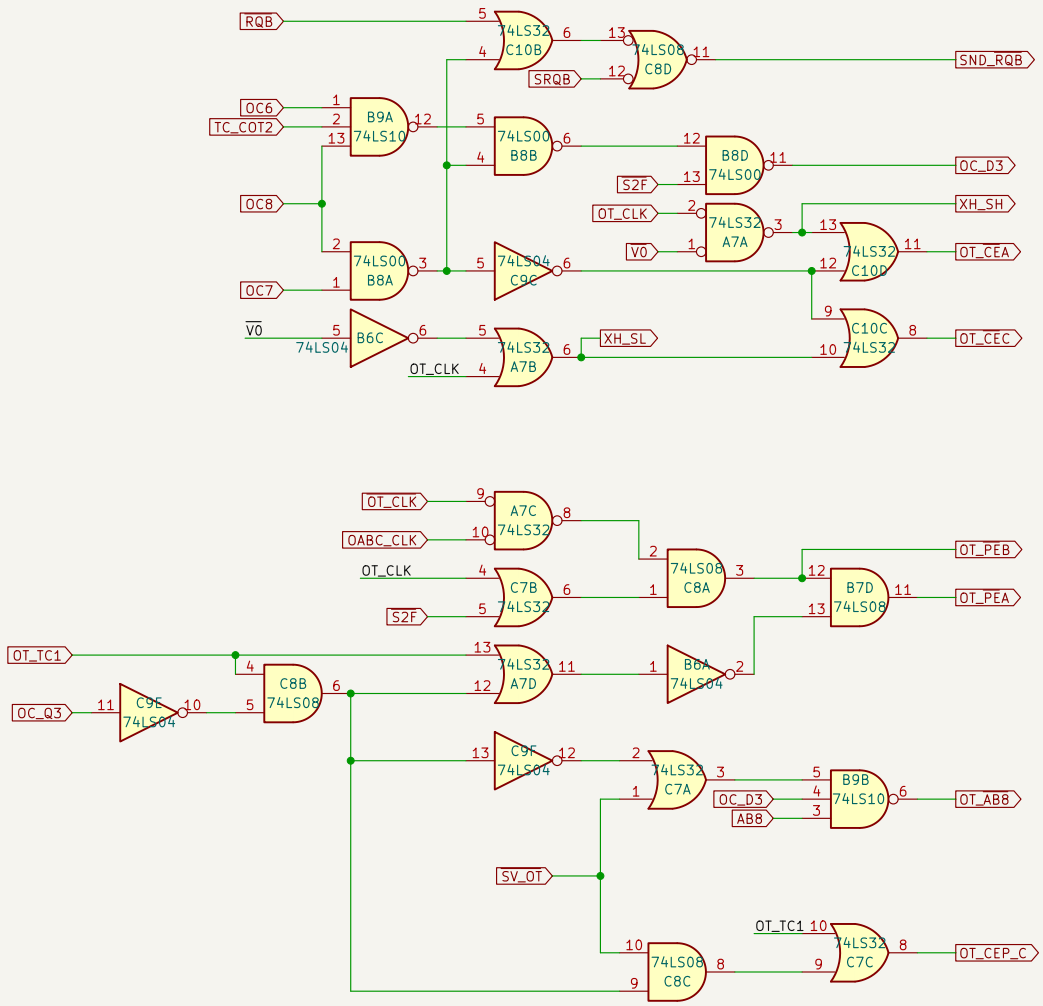
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JOTEGO

Sheet: /video/obj_ram/
File: obj_ram.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 19/28



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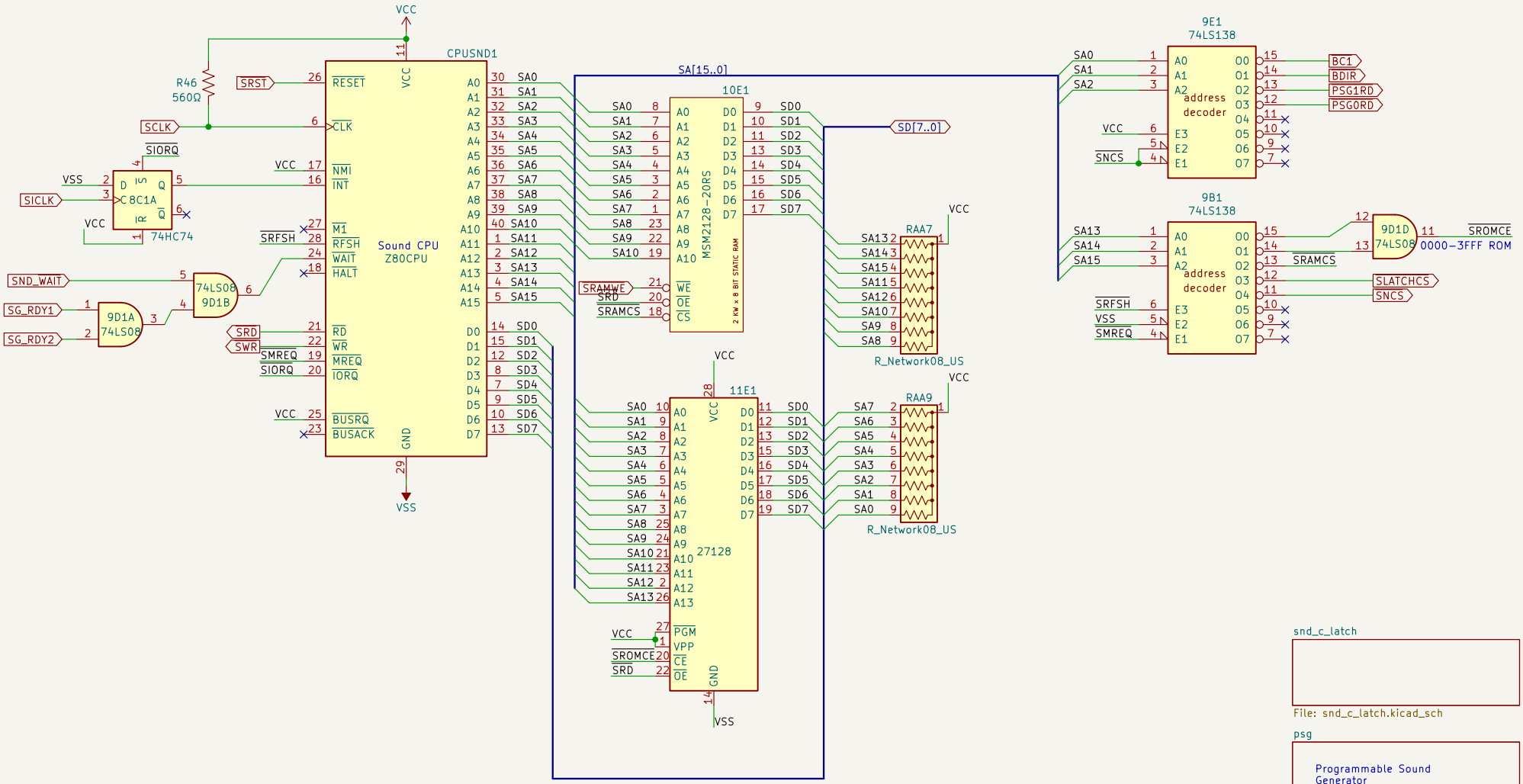
JOTEGO

Sheet: /video/logics_gates/

File: logics_gates.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 21/28



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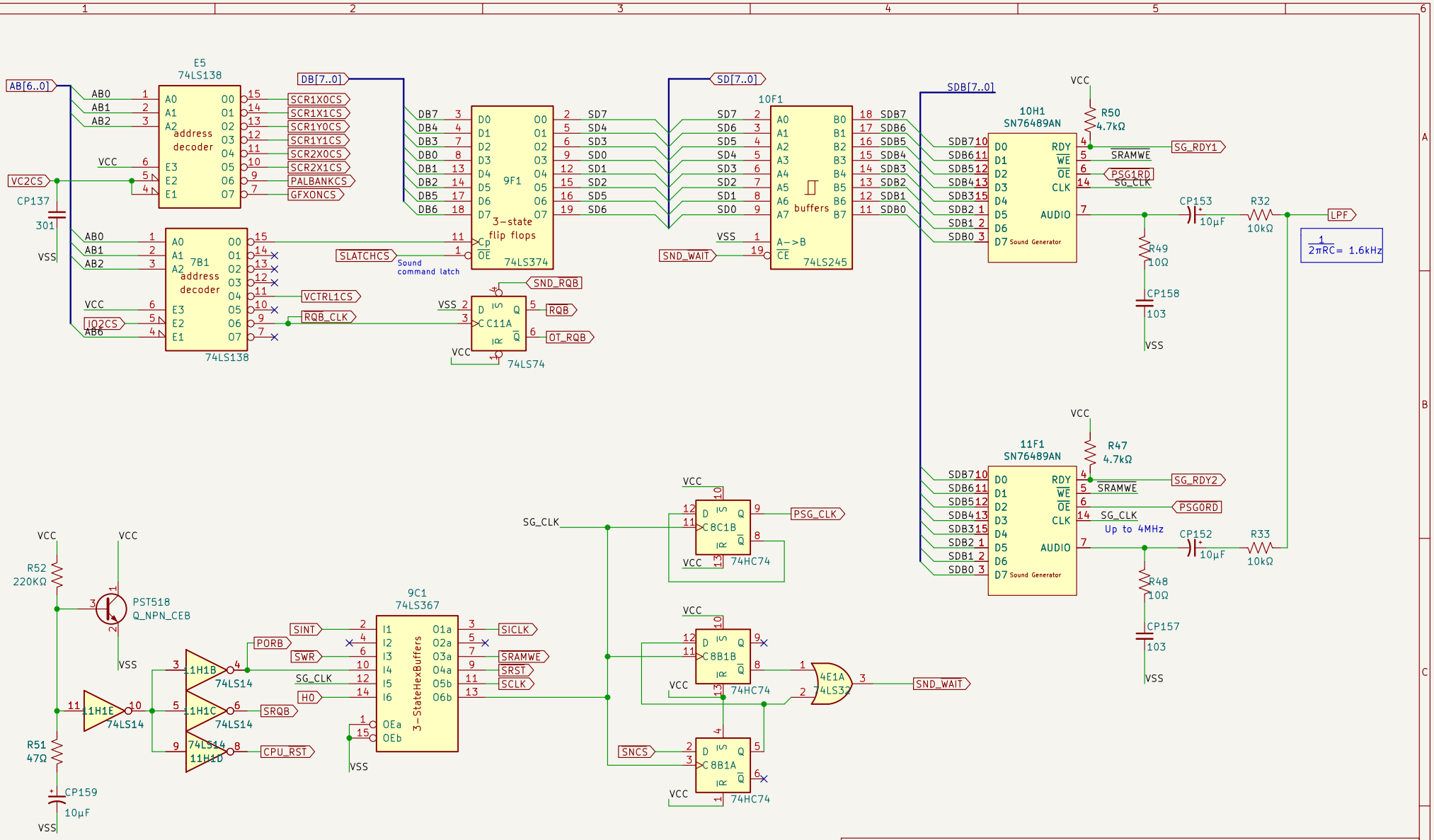
JOTEGO

Sheet: /sound/

File: sound.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 11/28



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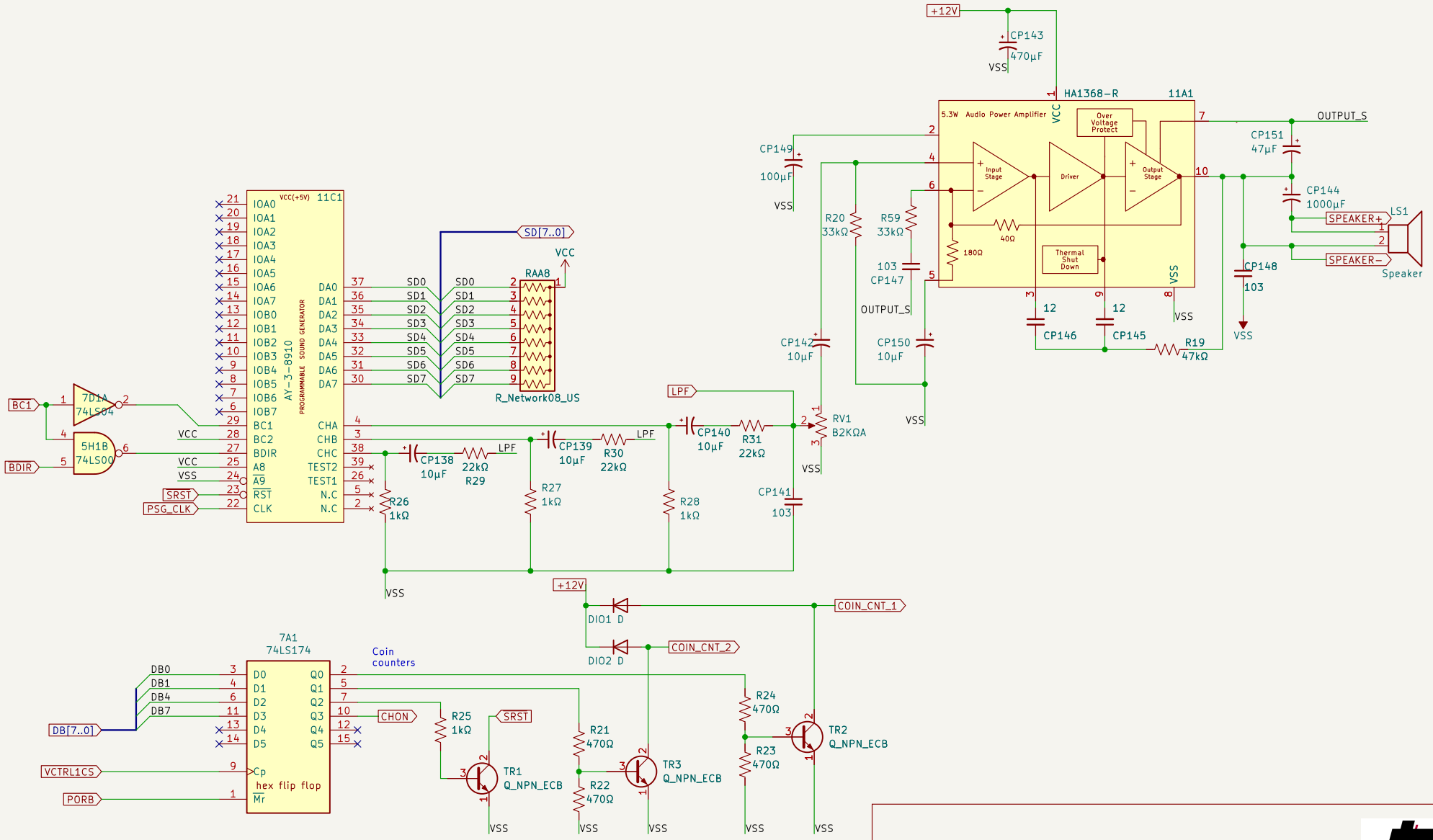
JOTEGO

Sheet: /sound/snd_c_latch/

File: snd_c_latch.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 12/28



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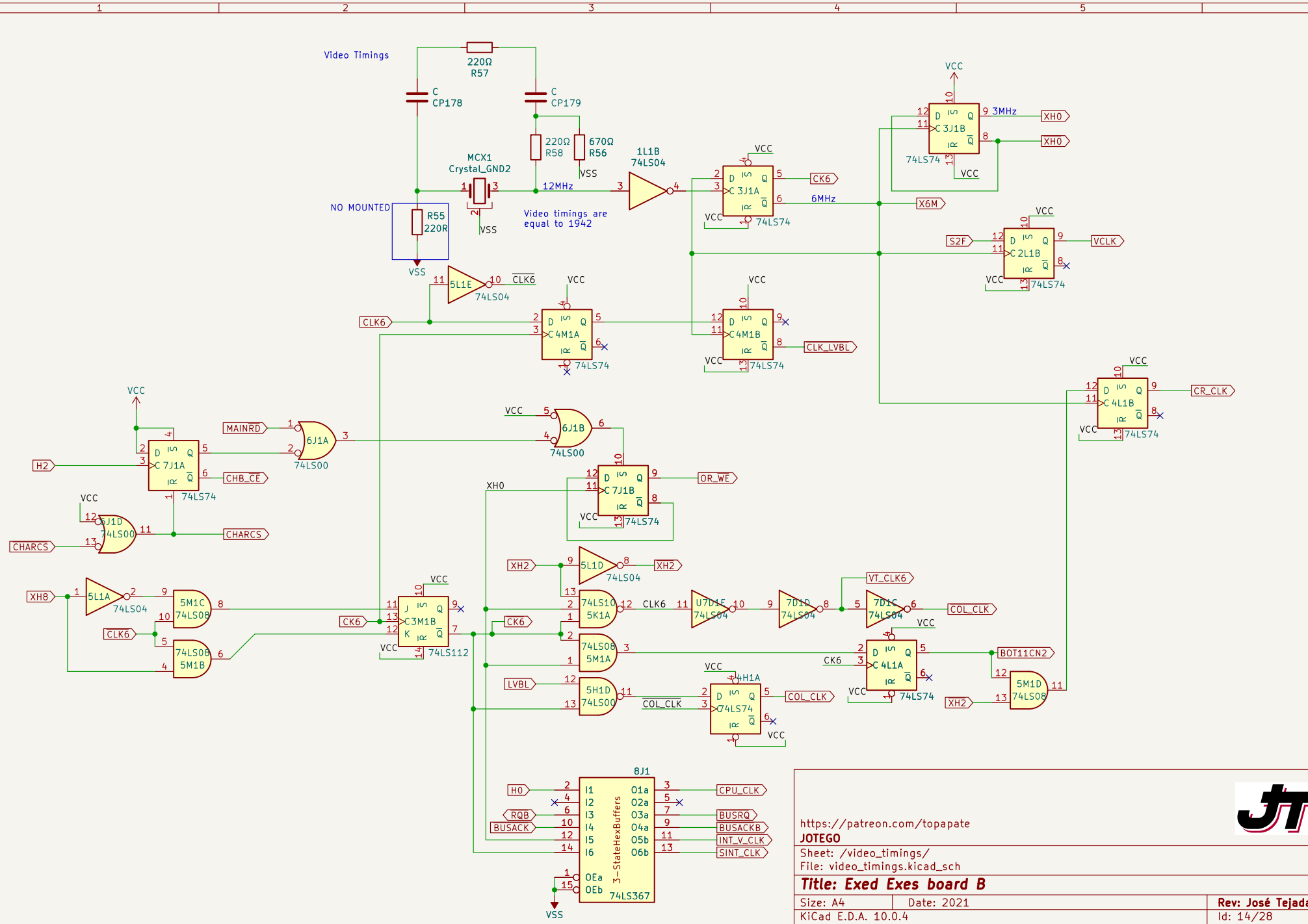
JOTEGO

Sheet: /sound/psg/

File: psg.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 26/28



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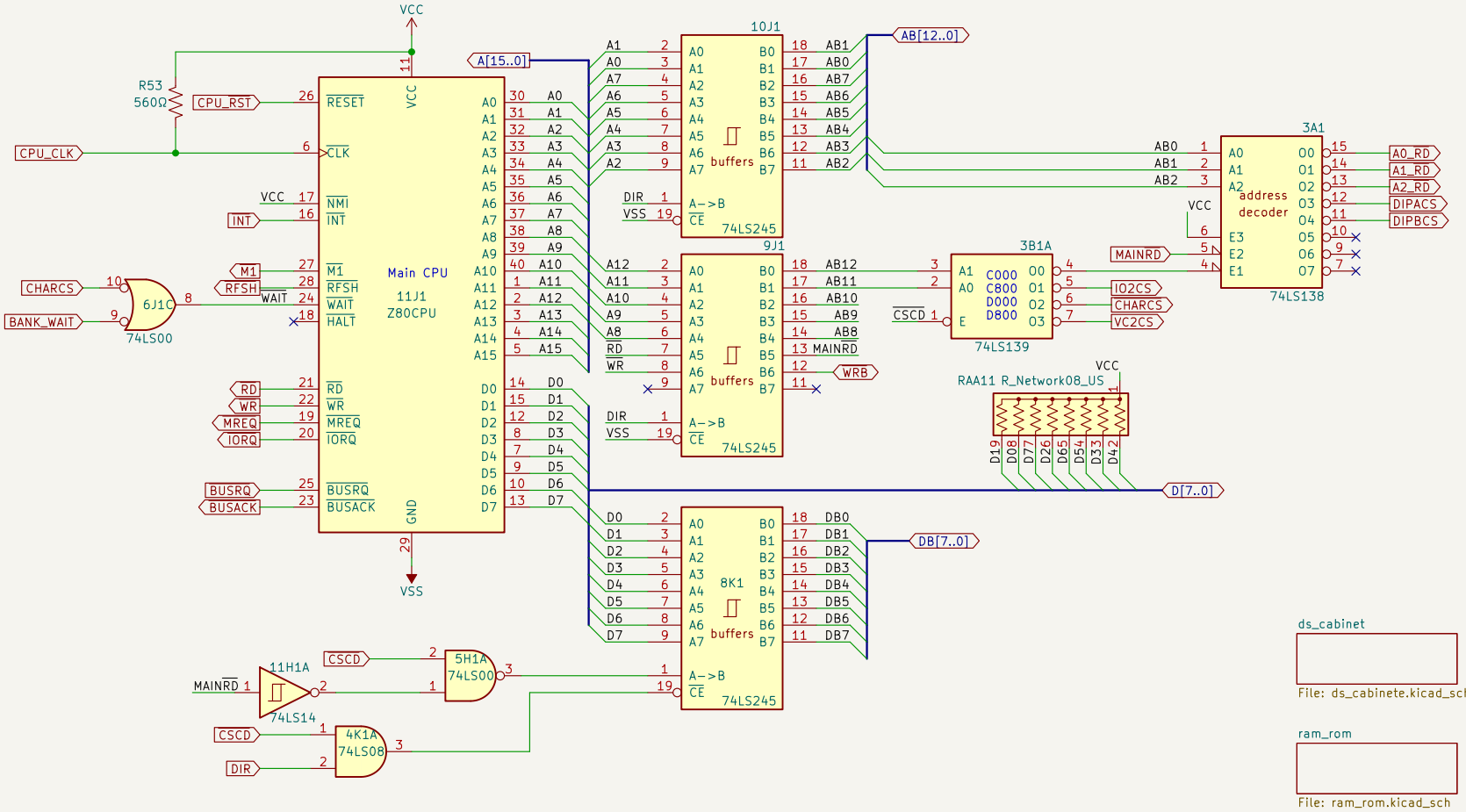
JOTEGO

Sheet: /video_timings/

File: video_timings.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 14/28



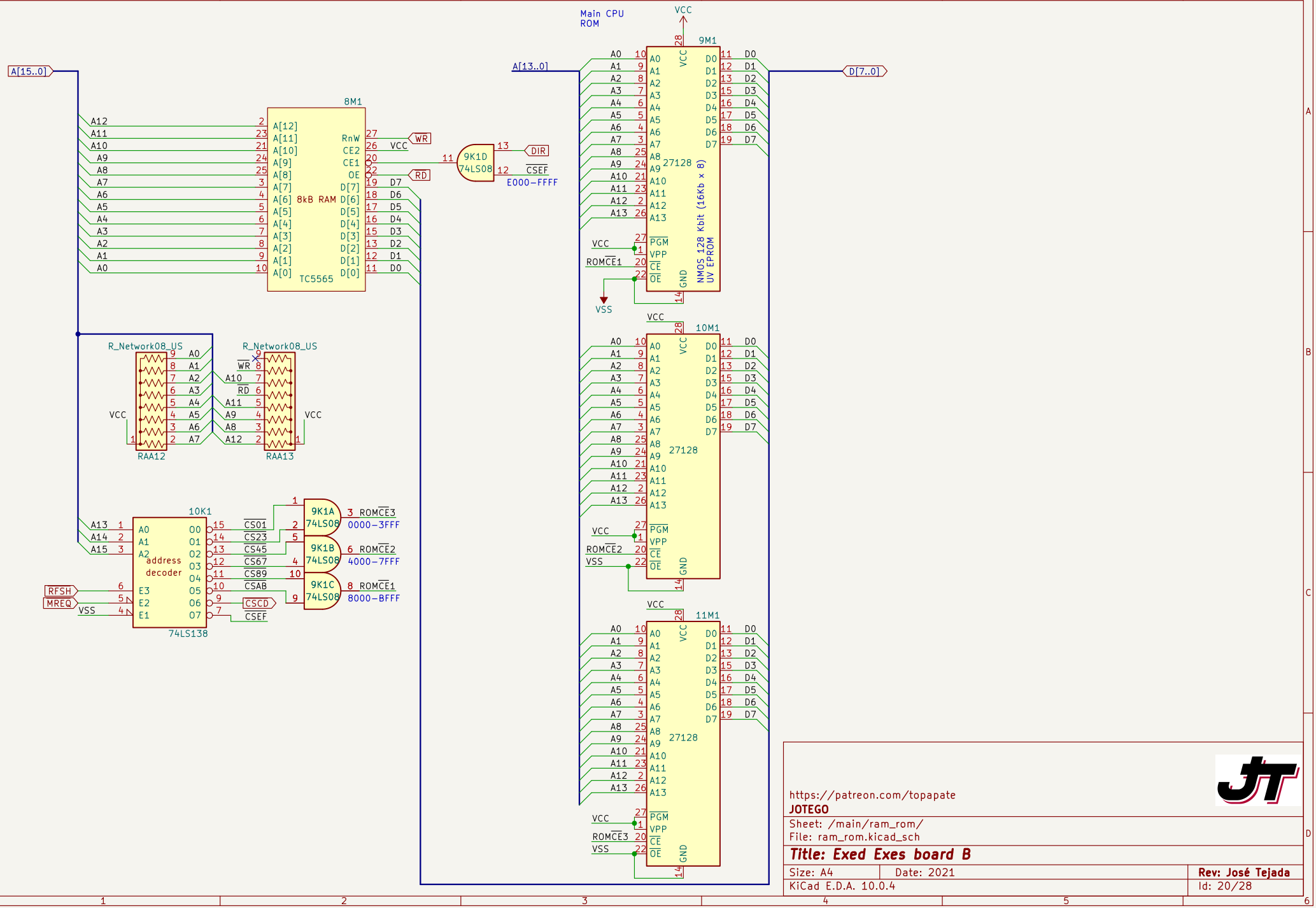
https://patreon.com/topapate

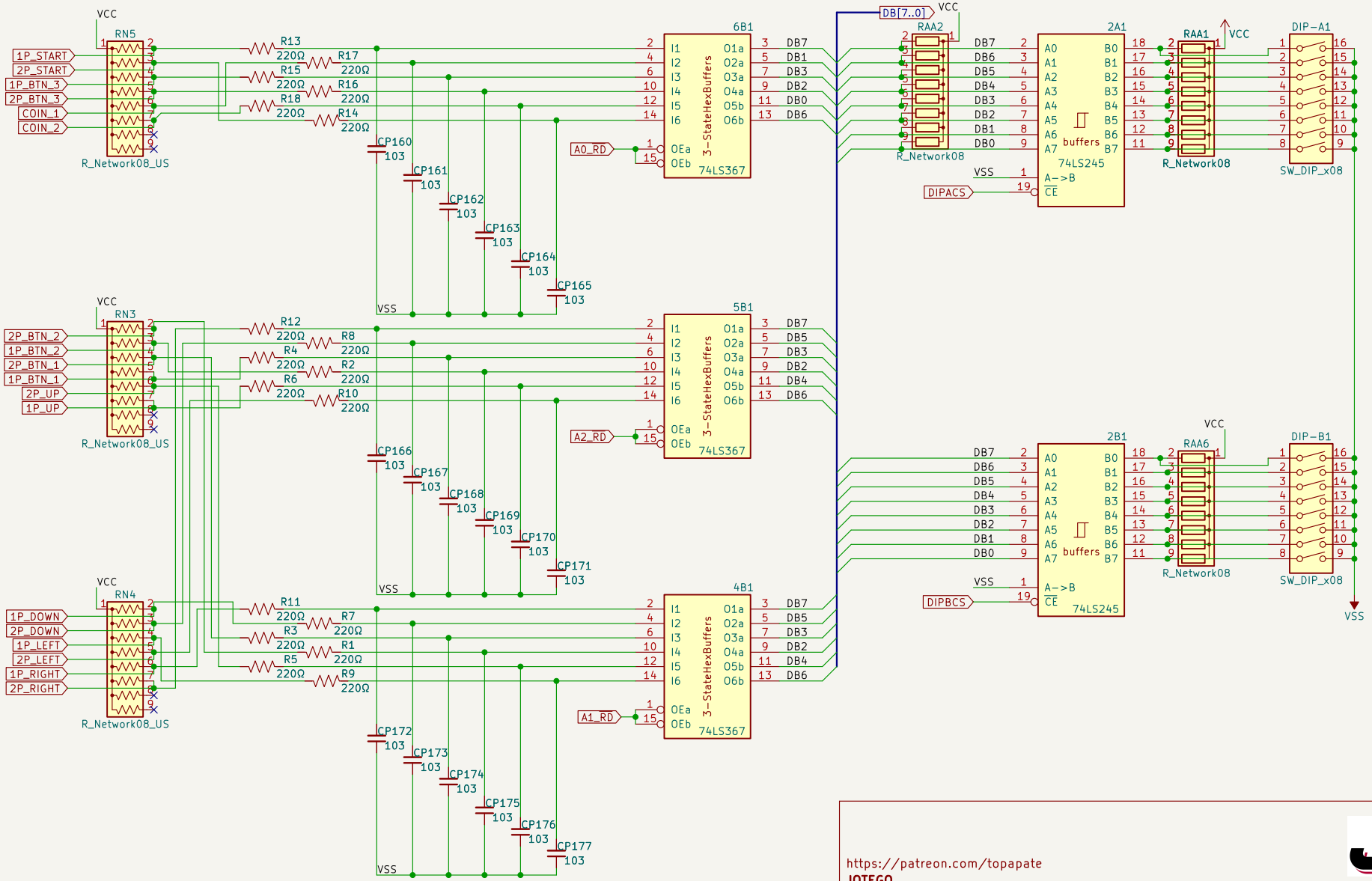
JOTEGO

Sheet: /main/
File: main.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 15/28


Main CPU
ROM
A[15..0]
A[13..0]
D[7..0]
8M1
TC5565
8kB RAM
WR
RD
9K1D
74LS08
DIR
CSEF
E000-FFFF
R_Network08_US
RAA12
RAA13
10K1
74LS138
address
decoder
RFSH
MREQ
9K1A
9K1B
9K1C
ROMCE3
0000-3FFF
ROMCE2
4000-7FFF
ROMCE1
8000-BFFF
9M1
10M1
11M1
27128
NMOS 128 Kbit (16Kb x 8)
UV EPROM
VCC
VSS
https://patreon.com/topapate
JOTEGO
Sheet: /main/ram_rom/
File: ram_rom.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 20/28



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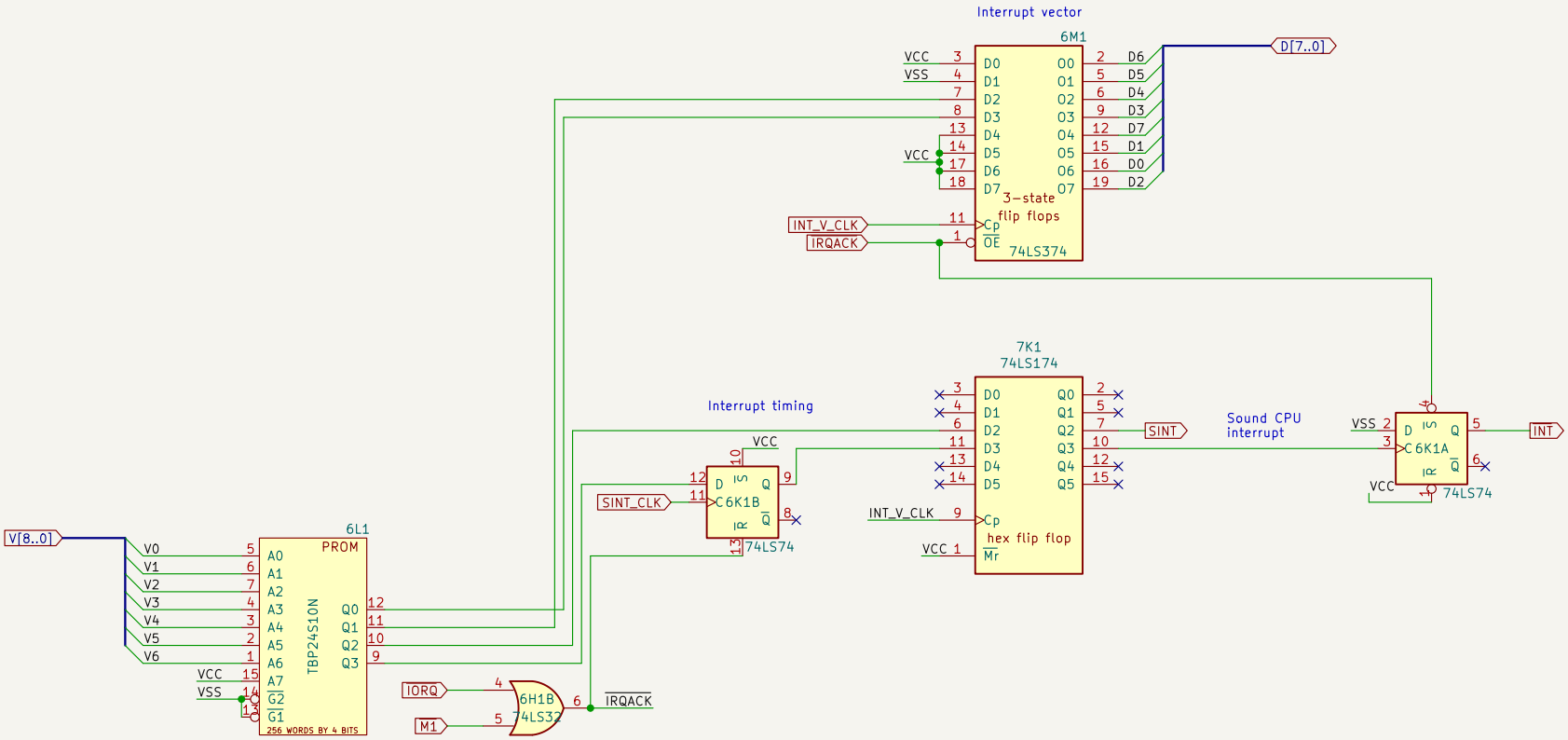
JOTEGO

Sheet: /main/ds_cabinet/

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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 26/28

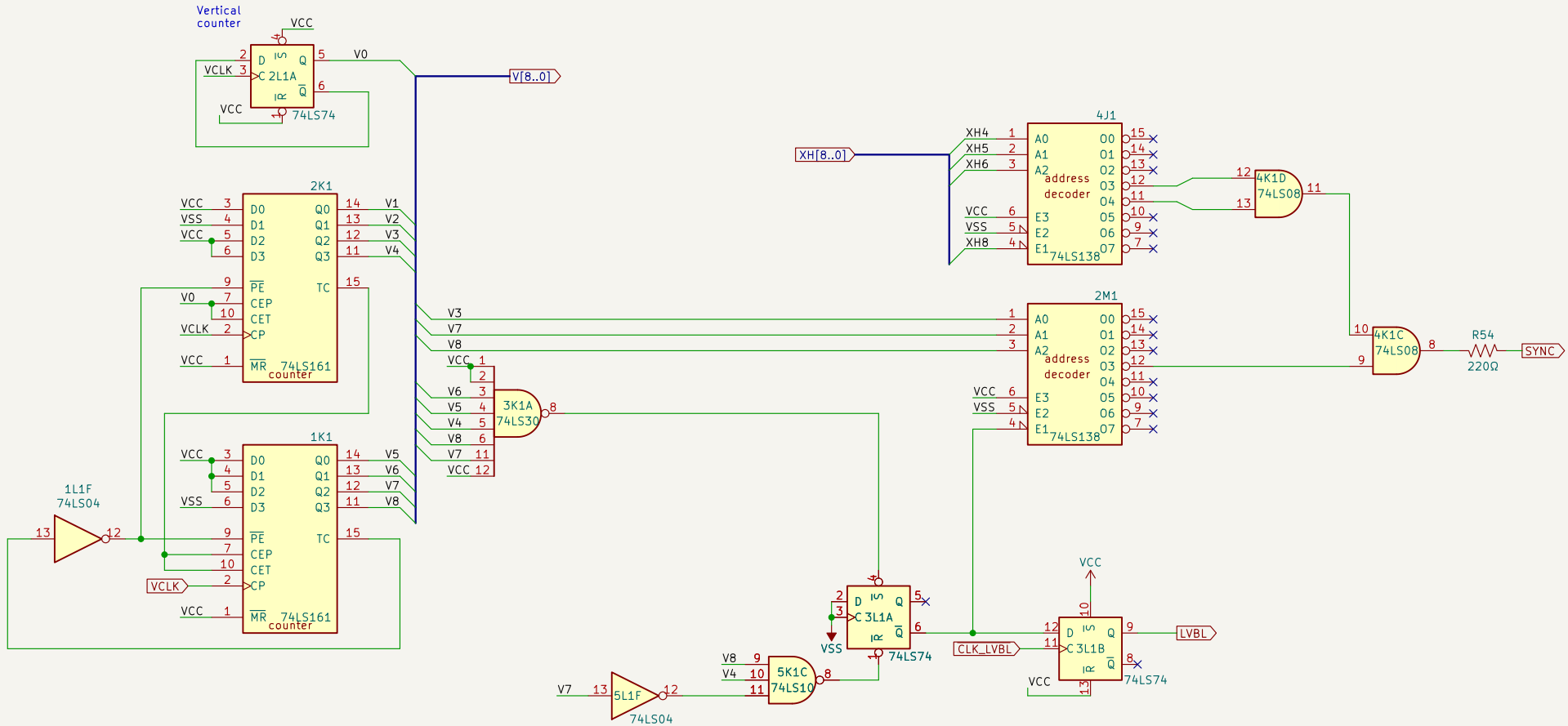


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JOTEGO

Sheet: /interruption/
File: interruption.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 16/28



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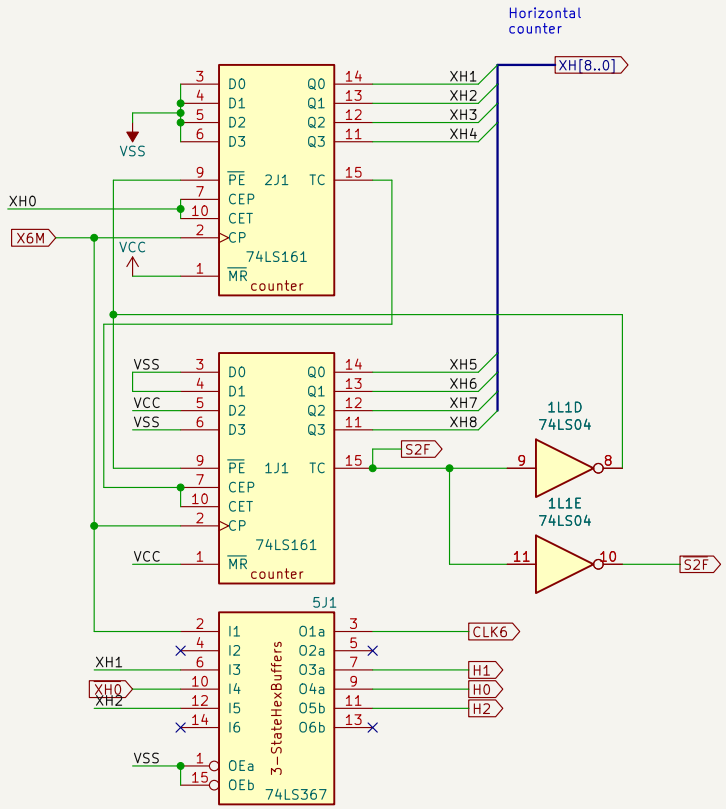
JOTEGO

Sheet: /v_counter/

File: v_counter.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 17/28



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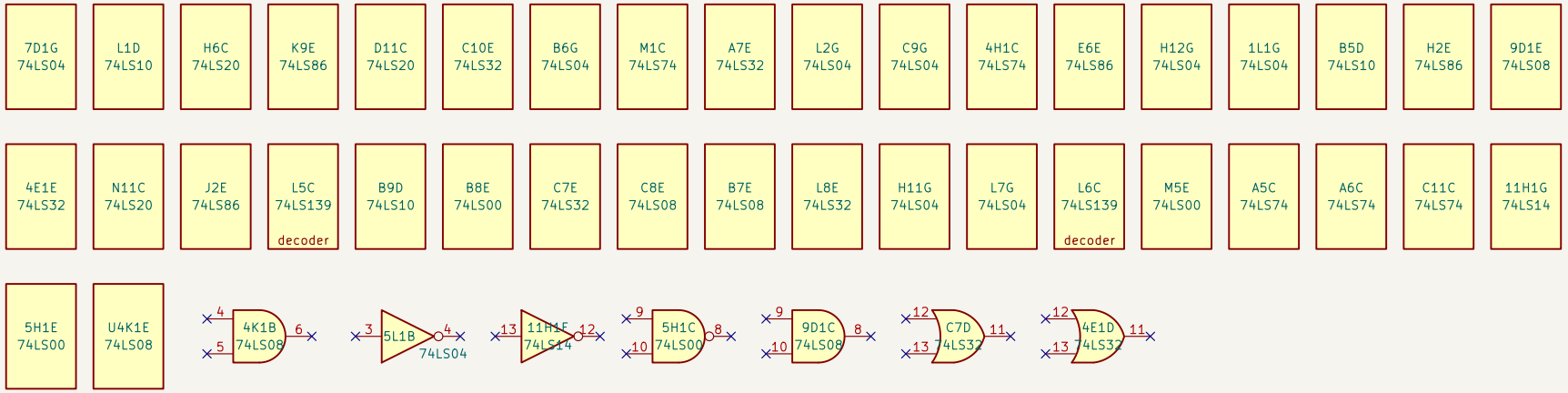
JOTEGO

Sheet: /h_counter/

File: h_counter.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 18/28



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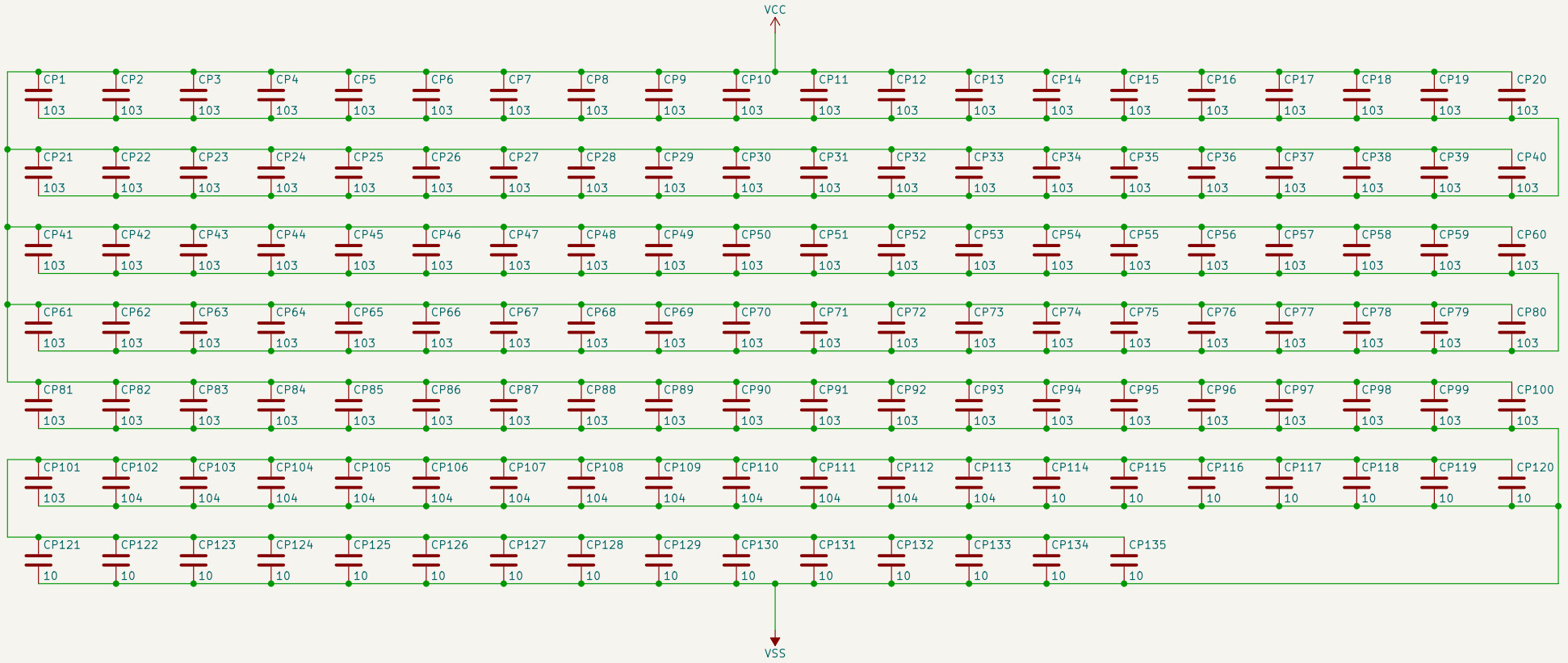
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Sheet: /last_u/

File: last_u.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 20/28



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Sheet: /capacitors/

File: capacitors.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 22/28